WBC TUTOR LMS — WATER SECTOR SKILLS DEVELOPMENT PROGRAMME

Introduction to Algebra

A Practical, Self-Paced Course for Confident, Real-World Problem Solving

Course Overview

Algebra is the everyday language of technical and vocational work — from calculating dosing rates and flow volumes to sizing tanks and interpreting formulas. This self-paced course builds algebra skills from the ground up, using clear explanations, fully worked examples, and — throughout — real water-sector scenarios that show exactly why each rule matters on the job. No prior algebra knowledge is assumed.

Who Is This Course For?

- Mid-secondary scholars (ages 16–18) looking to revise and strengthen their algebra before exams or further study.
- Water-sector technicians and operational staff undertaking upskilling or re-skilling as part of workplace development.
- Adult learners returning to study who want a practical, jargon-free refresher in core algebra.
- Anyone preparing for further technical or vocational training that assumes solid algebra fundamentals.

Course Purpose & Structure

This course takes learners on a carefully sequenced journey — from the basic rules of arithmetic and expressions, through factorising and solving equations, to confidently tackling real technical word problems. It is delivered entirely online via the WBC Tutor LMS, so learners can study at their own pace, revisit any topic at any time, and track their own progress.

Every one of the 13 topics follows the same consistent, easy-to-follow structure:

- Learning Objectives & Key Concepts — know exactly what you'll be able to do by the end of the topic.
- Definition & Explanation — plain-language teaching, with clearly labelled diagrams and figures.
- Key Mathematical Rules — the essential rules, numbered for quick reference.
- Worked Examples — step-by-step solutions, including realistic water-sector applications.
- Self-Study Practice — problems to attempt independently, with full worked solutions provided.
- Tips for Learners — practical advice to help you avoid the most common mistakes.
- Diagnostic Quiz — an auto-marked quiz covering every rule taught, so you can check your understanding before moving on.

A companion Quick Reference Sheet is also provided, summarising every rule from the course on a single page — ideal for ongoing use once the course is complete. Altogether, the course represents approximately 11–12 hours of self-paced study.

What You'll Learn — Topics & Objectives

The course progresses through 13 topics, building steadily from foundational skills to real-world problem solving:

No.	Topic	What You'll Learn
1	**Numbers, Signs & Order of Operations**	Apply BODMAS confidently to multi-step calculations involving negative numbers, powers and roots.
2	**Algebraic Language: Variables & Expressions**	Translate real-world scenarios into algebraic expressions, and identify terms, coefficients and constants.
3	**Simplifying Expressions**	Combine like terms and expand brackets accurately using the distributive law.
4	**Factorisation**	Apply four factorisation methods — common factor, difference of two squares, trinomials, and grouping.
5	**Fractions**	Add, subtract, multiply and divide fractions confidently, including real-world proportion problems.
6	**Exponents (Laws of Exponents)**	Apply the laws of exponents to simplify expressions and solve real-world scaling problems.
7	**Substitution & Evaluating Formulas**	Substitute given values into technical formulas and evaluate results correctly and in context.
8	**Solving Linear Equations (One Variable)**	Solve one-step and two-step equations using inverse operations, and check solutions by substitution.
9	**Solving Multi-Step Linear Equations**	Solve equations involving brackets, fractions, powers and roots; solve and represent inequalities.
10	**Solving Quadratic Equations**	Factorise and solve quadratic equations, and apply the quadratic formula when factorising isn't possible.
11	**Rearranging Formulas (Changing the Subject)**	Rearrange technical formulas to make a different variable the subject, without repeated substitution.
12	**Ratios, Proportions & Rates**	Solve direct and inverse proportion problems relevant to mixing, dosing and flow-rate calculations.
13	**Word Problems & Real-World Applications**	Apply a structured six-step process to translate and solve real technical and workplace problems.

Course Policies & Practical Information

1. Engagement with Facilitator

The learner / participant will be able to post questions and/or comments on the activities / tasks. The developer of this self-study course will regularly respond to the questions and comments. The developer may also arrange an online meeting to respond to the questions and/or comments. The date and time for such an online meeting, specific to a course, will be posted on the WBC LMS. All online meetings / forums, etc. may / will be recorded and posted on the WBC LMS for the benefit of all learners / participants registered for the course.

2. Flexibility / Accessibility of Course Material

The self-study courses are available 24/7 and the learner / participant can start at any time once registered!

3. Refunds

Cancelling and/or not completing the course does not qualify the learner / participant to be eligible for a refund. Refunds will only be considered if there are compelling reasons (for cancellation / termination) beyond the control of the learner / participant.

4. In Addition

- The Foundation Courses are not related to any qualification but may serve to support specific content of a related qualification.
- The Foundation Courses are not associated with specific NQF levels.
- This Foundation Courses are not accredited by the Engineering Council of South Africa (ECSA) and/or the South African Council for Natural Scientific Professions (SACNASP).

5. Course Fees

The introductory fee for the online **Introduction to Algebra** self-study course **for employed staff is R 250**.

The **significantly reduced introductory fee** for the online **Introduction to Algebra** self-study course **for registered (current registrations) full-time and part-time scholars** at any education institution is **R 50**. Scholars will be required to provide a copy of their annual registration form (current year) issued by the educational institution. Please submit your documentation to: students@waterbusinesscollege.co.za.

Ready to build your algebra confidence?

Enrol today through the WBC Tutor LMS and start learning at your own pace.

Please note: For this pilot phase only your full names & email address is required